

Computational Optical System Modeling and Image Reconstruction

A physically validated Fourier-optics simulator:
pupil $\rightarrow$ PSF $\rightarrow$ MTF $\rightarrow$ detector $\rightarrow$ deconvolution

Abstract

This report describes a reproducible simulator for diffraction-limited incoherent imaging, written directly on NumPy and validated against closed-form diffraction theory rather than against its own plausibility. All 21 automated physics checks pass: the simulated modulation transfer function of a circular pupil agrees with the analytic incoherent form to an RMS residual of 2.95×10^{-4} , the measured point-spread-function full width at half maximum reproduces the Airy value $1.02899 \lambda/2\text{NA}$ to 1.02947, and two structurally independent routes to the optical transfer function agree to 4.65×10^{-16} . On that foundation the report measures three things that a purely visual study cannot. First, the Maréchal approximation for the Strehl ratio, quoted everywhere as though it were general, overstates the true value by a factor of 3.7 at a quarter wave RMS of defocus (0.0848 against 0.0229). Second, the regularization parameter of every deconvolution method matters more than the choice of method, and the parameter that maximizes PSNR is not the one that maximizes SSIM — at one noise level the best PSNR is 27.89 dB from Wiener (spectral K) while the best SSIM, 0.751, comes from Wiener (constant K) at a setting whose PSNR is lower. Third, detector undersampling is a corruption rather than a loss: at 4.0 times the Nyquist pitch, 36.7% of a broadband target's modulation reappears at the wrong frequency, and the frequency at which it reappears matches the folding prediction to 3.49×10^{-10} cycles per metre.

1 Scope and questions

The object of study is the classical imaging chain, in full and end to end:

$$g(x, y) = [f * \text{PSF}](x, y) \xrightarrow{\text{detector}} \text{sampld} + n(x, y),$$

with the PSF generated from a complex pupil function that may carry Zernike aberrations, the detector modelled as area integration followed by sampling, and the noise either additive Gaussian or Poisson. The questions the report answers are the ones a working optical scientist actually asks: how do wavelength and numerical aperture set resolution; what does a given amount of aberration cost in measurable contrast; when does deconvolution help and when does it

only amplify noise; and how coarse can a detector be before its sampling starts inventing structure.

No machine learning is used, and no external optics library is used in the forward model. `scikit-image` supplies SSIM and the standard test photographs; everything optical is implemented here, which is what makes the validation section meaningful rather than circular.

2 Method

Units and grids. SI throughout. Lengths are metres, so spatial frequencies are cycles per metre. Two dimensionless quantities recur: the normalized pupil radius $\rho = f_r/(\text{NA}/\lambda)$, unity at the pupil rim, and the normalized spatial frequency $\nu = f/f_c$ with $f_c = 2\text{NA}/\lambda$ the incoherent cutoff. Zernike coefficients are in waves RMS.

Every array is stored *centred*, with the origin at index $n/2$. The two Fourier helpers absorb the `fftshift/ifftshift` pair and carry the dx^2 and df^2 weights that make them exact inverses in physical units; no other module in the package calls `fftshift` on a field. Centring errors are the dominant failure mode in Fourier-optics code and are confined here to one file.

Where the pupil lives. The pupil is defined on the *image-plane frequency grid*, with its rim at the coherent cutoff NA/λ , rather than on a separate pupil-plane length grid. No focal length appears anywhere, because it cancels out of every image-space quantity once the system is described by its NA. The consequence that matters is that the incoherent cutoff $f_c = 2\text{NA}/\lambda$ is not a number typed into the code: the OTF is the autocorrelation of a function of radius NA/λ , so its support has radius $2\text{NA}/\lambda$ automatically. The cutoff check in Table 1 is therefore a real test.

The chain. From the generalized pupil $P = A \exp(i\phi)$ with $\phi = 2\pi W$ and W a sum of Noll-normalized Zernike modes, the amplitude PSF is $h_A = \mathcal{F}^{-1}\{P\}$, the intensity PSF is $|h_A|^2$ normalized to unit sum, the OTF is $\mathcal{F}\{\text{PSF}\}$ normalized to unity at zero frequency, and the MTF is its modulus. Unit-sum normalization makes the PSF an energy-conserving convolution kernel, so any change in mean brightness in a simulated image is deliberate.

Sampling. The simulation grid is set from the optics: the Nyquist pitch for the incoherent cutoff is $\lambda/(4\text{NA})$, and the simulation runs at half of that. Detector undersampling is then modelled by resampling onto a coarse detector, never by coarsening the pupil grid, so the optics stay correctly sampled while the detector is deliberately not. Detector pitch is always expressed as a multiple of the Nyquist *pitch*; a factor above one undersamples.

Restoration. Three methods. The naive inverse divides by the OTF wherever $|H|$ exceeds a floor. The Wiener filter is $\hat{F} = H^*G/(|H|^2 + K)$, in a constant- K form and in a form where $K(f)$ is estimated from the measurement itself. Richardson–Lucy is the multiplicative Poisson maximum-likelihood iteration, implemented from its update equation.

One point of algebra is worth stating because it is easy to miss and would otherwise produce a meaningless comparison: the “regularized inverse” $\hat{F} = H^*G/(|H|^2 + \epsilon)$ is the Wiener filter with $K = \epsilon$. They are not two methods. The naive inverse reported here is therefore the genuine division it is meant to be contrasted with.

3 Validation

Table 1 is the foundation for everything after it. Three of its rows deserve comment.

The MTF row compares the simulated radial MTF against $\text{MTF}(\nu) = \frac{2}{\pi}[\arccos \nu - \nu\sqrt{1 - \nu^2}]$ and reports an *absolute* RMS residual of 2.95×10^{-4} over $\nu \leq 0.95$. The comparison stops short of the cutoff deliberately: the last few percent are where a pixelated pupil rim stops being a circle, and including them would report a property of the rim rather than of the method. Figure 1 plots the residual so the reader can see where it grows instead of taking the single number on trust; its peak is 1.53×10^{-3} .

The autocorrelation row compares two structurally different computations of the same object — $\mathcal{F}\{|\mathcal{F}^{-1}\{P\}|^2\}$ against the direct autocorrelation of P — and finds 4.65×10^{-16} . Since the two share almost no code path, this is a genuine check on the transform conventions rather than a restatement of them.

The grating rows close the loop differently again: they form an image of a sinusoid of known frequency and remeasure its contrast, which is the *operational* definition of the MTF, and compare that with the analytic curve. Agreement is better than 0.5% absolute at every frequency tested.

4 Diffraction-limited performance

Table 2 sweeps four wavelengths against five numerical apertures. Read down the normalized columns rather than the absolute ones: the FWHM expressed in units of λ/NA varies only between 0.514734 and

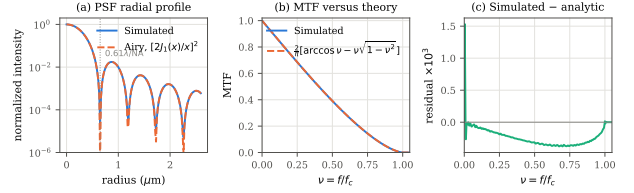


Figure 1: Simulation against closed-form theory. (a) PSF radial profile versus the Airy pattern over six decades. (b) Radial MTF versus the analytic incoherent MTF. (c) The residual, plotted separately because at the scale of (b) a 0.1% error is invisible.

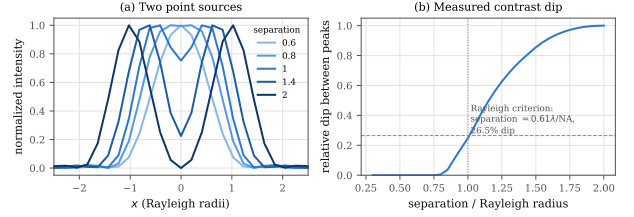


Figure 2: Two point sources at increasing separation, and the contrast dip between their peaks. At exactly one Rayleigh radius the measured dip is 0.2475, against the textbook 0.2644; the difference is not numerical error but a difference in definition, discussed in the text.

0.514815 across all 20 systems, against the theoretical 0.514494, and MTF_{50}/f_c sits at 0.4037 against the analytic 0.4040.

An honest caveat applies to that collapse, and Fig. 3 states it on the figure itself. The simulation grid is *defined* in units of λ/NA , so all eight rescaled profiles are the same computation and agree to machine precision. The collapse confirms that the implementation carries its units consistently; it is not independent evidence for the scaling law. The evidence for the physics is the comparison against the analytic curves in Fig. 1.

At the baseline system (532 nm, NA 0.5) the cutoff is 1.8797 cycles per micrometre, the Abbe limit $\lambda/2\text{NA}$ is 532.0 nm, the Rayleigh radius $0.61\lambda/\text{NA}$ is 649.0 nm, and the measured PSF FWHM is 547.7 nm. Those three “resolutions” differ by up to 20%, which is why the tables carry all of them.

Figure 2 treats the Rayleigh criterion as an experiment rather than a definition. The two sources are placed by an exact Fourier-domain shift, not by rounding two delta functions onto the pixel grid — at this scale that rounding is a 0.2-Rayleigh error, large enough to move the apparent resolution limit for a purely numerical reason. The measured dip at one Rayleigh radius is 0.2475 against the quoted 0.2644. The gap is a definitional one: the textbook figure evaluates the intensity at the two source positions, whereas the summed pattern’s actual local maxima sit slightly further apart and are slightly dimmer. Reporting 0.2475 and explaining it is more useful than tuning the measurement until it returns 0.2644.

Table 1: Physics validation. Every check is against a closed-form result, not against a previous run of this code. Tolerances are set by discretization, and a row that failed would be printed as failing. 21 of 21 pass.

Quantity	Expected	Measured	Rel. error	Tolerance	
PSF sum (normalized)	1.0000	1.0000	1.11×10^{-16}	1.00×10^{-10}	✓
MTF(0)	1.0000	1.0000	1.11×10^{-16}	1.00×10^{-10}	✓
PSF FWHM / ($\lambda/2\text{NA}$)	1.0290	1.0295	4.68×10^{-4}	2.00×10^{-2}	✓
Encircled-energy radius / Rayleigh radius	1.0000	1.0240	2.40×10^{-2}	5.00×10^{-2}	✓
RMS(MTF simulated - analytic), $\nu_j = 0.95$	0.0000	0.0003	2.95×10^{-4}	2.00×10^{-3}	✓
MTF50 / f_c	0.4040	0.4037	5.80×10^{-4}	1.00×10^{-2}	✓
MTF10 / f_c	0.8054	0.8049	5.71×10^{-4}	1.00×10^{-2}	✓
Cutoff (1% MTF) / f_c	0.9587	0.9582	5.57×10^{-4}	2.00×10^{-2}	✓
max —OTF(autocorrelation) - OTF(FFT of PSF)—	0.0000	0.0000	4.65×10^{-16}	1.00×10^{-10}	✓
Imaged grating contrast at $\nu=0.250$	0.6850	0.6852	2.51×10^{-4}	2.00×10^{-2}	✓
Imaged grating contrast at $\nu=0.500$	0.3910	0.3908	5.08×10^{-4}	2.00×10^{-2}	✓
Imaged grating contrast at $\nu=0.750$	0.1443	0.1439	3.02×10^{-3}	2.00×10^{-2}	✓
max —PSF(zero Zernikes) - PSF(ideal)—	0.0000	0.0000	0	1.00×10^{-15}	✓
Strehl (PSF) vs Strehl (pupil integral), 0.05 waves RMS	0.9053	0.9053	3.68×10^{-16}	1.00×10^{-6}	✓
Strehl (PSF) vs Strehl (pupil integral), 0.1 waves RMS	0.6632	0.6632	5.02×10^{-16}	1.00×10^{-6}	✓
Strehl (PSF) vs Strehl (pupil integral), 0.2 waves RMS	0.1435	0.1435	0	1.00×10^{-6}	✓
Strehl decreases monotonically with defocus	1.0000	1.0000	0	0	✓
Aliased modulation fraction, detector pitch = Nyquist pitch	0.0000	0.0000	1.34×10^{-30}	1.00×10^{-6}	✓
Aliased modulation fraction, detector pitch = 2x Nyquist pitch	1.0000	1.0000	0	1.00×10^{-6}	✓
Apparent frequency of an aliased grating	3.82×10^5	3.82×10^5	3.05×10^{-16}	2.00×10^{-2}	✓
max —Fraunhofer(circular) - analytic Airy—	0.0000	0.0002	1.54×10^{-4}	2.00×10^{-3}	✓

5 Aberrations

Table 3 sweeps defocus, astigmatism, coma and spherical aberration over six severities each. Severity is in waves *RMS* of a single Noll-normalized mode; the peak-to-valley equivalent is a different number for each mode (3.464 for defocus, 4.899 for astigmatism) and is carried as its own column, because half the optics literature quotes PV and a factor of three-and-a-half is exactly the kind of discrepancy that makes two aberration studies appear to disagree.

Two results stand out. The first is quantitative and negative: the Maréchal approximation $S \approx e^{-(2\pi\sigma)^2}$ tracks the true Strehl closely at a tenth of a wave (0.6738 against 0.6632) and then fails, overstating it by a factor of 3.7 at a quarter wave (0.0848 against 0.0229) and by orders of magnitude beyond that. Figure 4(b) shows where the two curves part company. The approximation is a small-aberration expansion and behaves like one; quoting it outside the diffraction-limited regime, as is common, is not conservative.

The second is that the Strehl ratio and the MTF disagree about which aberration is worst, because they measure different things. At one wave RMS, Spherical aberration has the lowest MTF50 (0.0149 f_c) while retaining a comparatively high on-axis Strehl — it redistributes energy into rings without hollowing the core, so the peak survives while mid-frequency contrast does not. A single scalar quality metric cannot capture that, which is the argument for reporting the full MTF curve (Fig. 5) alongside it. Already at a tenth of a wave RMS of defocus — an aberration a Strehl of 0.6632 would call “nearly diffraction limited” — MTF50 has fallen by 39.9%.

6 Noise and restoration

Table 4 compares five treatments across additive Gaussian noise and Poisson shot noise, and Fig. 7 plots both metrics against both noise families.

The ordering changes with noise, and the reason is structural rather than incidental. At zero noise the best method is Wiener (constant K) at 30.36 dB; at $\sigma = 0.1$ it is Wiener (spectral K) at 24.21 dB. At $\sigma = 0.05$, doing nothing scores 23.10 dB and SSIM 0.336, the naive inverse makes matters actively worse (21.43 dB, SSIM 0.284), Richardson–Lucy at a fixed 20 iterations barely holds its ground (22.48 dB), and only the spectrally weighted Wiener filter improves both metrics substantially (26.41 dB, SSIM 0.660).

Regularization dominates method choice

Table 5 sweeps each method across its own regularization parameter at one noise level. The spread *within* a method is larger than the spread between methods: the naive inverse ranges from 7.24 dB at a weak floor — far below the 25.44 dB of doing nothing — up to a value competitive with everything else once the floor is raised enough that little deconvolution is actually happening. That is the real lesson of inverse filtering, and it is visible only because the parameter was swept rather than fixed at a plausible-looking value.

The two metrics also disagree about the answer. The best PSNR is 27.89 dB from Wiener (spectral K); the best SSIM is 0.751 from Wiener (constant K) at $K=0.1$, a setting that regularizes harder and scores worse on PSNR. PSNR rewards suppressing noise energy; SSIM rewards preserving local structure. Neither is wrong, and a study that reported only one would have concluded something different about which setting to use.

Table 2: Optical performance over the wavelength $\times$ NA matrix. Read the normalized columns: they should be constant down the whole table, and their spread is discretization error rather than physics. NA 1.0 is paired with a water-immersion index, because $\text{NA} = n \sin \theta$ cannot exceed n .

λ (nm)	NA	FWHM (nm)	FWHM theory (nm)	f_c (cyc/ μm)	MTF50/ f_c	MTF10/ f_c	cutoff/ f_c
450	0.1	2316.3	2315.2	0.444	0.4037	0.8049	0.9582
450	0.25	926.5	926.1	1.111	0.4037	0.8049	0.9582
450	0.5	463.3	463.0	2.222	0.4037	0.8049	0.9582
450	0.75	308.8	308.7	3.333	0.4037	0.8049	0.9582
450	1	231.6	231.5	4.444	0.4037	0.8049	0.9582
532	0.1	2738.4	2737.1	0.376	0.4037	0.8049	0.9582
532	0.25	1095.4	1094.8	0.940	0.4037	0.8049	0.9582
532	0.5	547.7	547.4	1.880	0.4037	0.8049	0.9582
532	0.75	365.1	364.9	2.820	0.4037	0.8049	0.9582
532	1	273.8	273.7	3.759	0.4037	0.8049	0.9582
633	0.1	3258.3	3256.7	0.316	0.4037	0.8049	0.9582
633	0.25	1303.3	1302.7	0.790	0.4037	0.8049	0.9582
633	0.5	651.7	651.3	1.580	0.4037	0.8049	0.9582
633	0.75	434.4	434.2	2.370	0.4037	0.8049	0.9582
633	1	325.8	325.7	3.160	0.4037	0.8049	0.9582
850	0.1	4375.2	4373.2	0.235	0.4037	0.8049	0.9582
850	0.25	1750.1	1749.3	0.588	0.4037	0.8049	0.9582
850	0.5	875.0	874.6	1.176	0.4037	0.8049	0.9582
850	0.75	583.5	583.1	1.765	0.4037	0.8048	0.9580
850	1	437.5	437.3	2.353	0.4037	0.8049	0.9582

Iteration count is a regularization parameter

Richardson–Lucy has no explicit regularizer, so the iteration count becomes one. Table 6 and Fig. 8 make the point sharply: with no noise the metrics are still improving at 60 iterations (60); at $\sigma = 0.01$ PSNR peaks at iteration 34 but SSIM peaks at 9; at $\sigma = 0.05$ both peak at iteration 3. A fixed recommendation of “20 iterations” is therefore wrong in both directions depending on the noise, and by a factor of ten. The cost is not negligible either: at the largest iteration count Richardson–Lucy is roughly 55 times slower than a Wiener filter on the same image.

6.1 What it costs not to know the PSF

Every number above assumes the restoration is handed the exact PSF that degraded the image. Real deconvolution is not given that. Table 7 degrades with a quarter wave of defocus and restores with a different assumed defocus. With the correct PSF the Wiener filter reaches 22.85 dB; with the assumed defocus wrong by 0.25 waves it falls to 20.69 dB. Richardson–Lucy behaves similarly (22.51 dB to 21.18 dB). The advantage of a good restoration is real but it is not robust, which is the honest context for every oracle-PSF number in this report.

7 Detector sampling

The optics band-limit the image at f_c ; the detector then does two separate things, and they are routinely conflated. Pixel area integration is a convolution with

a box of width p , transfer function $|\text{sinc}(pf)|$ — a real and irreversible loss of contrast. Sampling replicates the spectrum every $1/p$, and if $1/p < 2f_c$ the replicas overlap. That second effect is not a loss of contrast but a corruption: the energy is still present, sitting at the wrong frequency and indistinguishable from signal.

Table 8 shows how strongly the answer depends on what is being imaged. At 4.0 times the Nyquist pitch, a broadband chirp target loses 36.7% of its modulation to folding while a photograph loses 1.1% — not because the detector behaves differently, but because a photograph has little energy near the optical cutoff to fold in the first place. A sampling study run only on photographs would substantially understate the risk.

Figure 10 shows the folding law directly. Gratings above the detector’s Nyquist frequency reappear at $|f_s - f|$, and across 6 such gratings the measured apparent frequency matches the prediction to 3.49×10^{-10} cycles per metre. The important feature of that figure is the symmetry: once a grating has folded, nothing in the sampled image says which side of the Nyquist frequency it came from.

8 Key findings

1. The simulator reproduces closed-form diffraction theory to a few parts in 10^4 across 21 independent checks, including two structurally different routes to the OTF that agree to machine precision.
2. The Maréchal Strehl approximation overstates the true Strehl by $\times 3.7$ at a quarter wave RMS and diverges beyond that. It should not be quoted

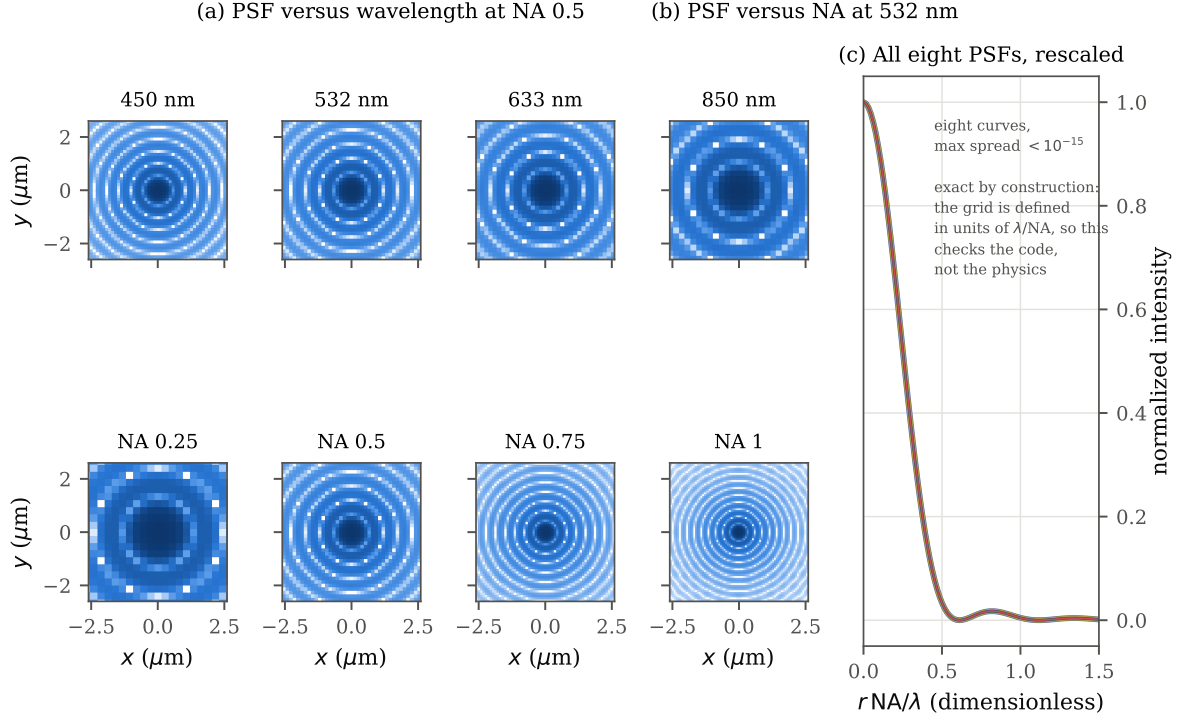


Figure 3: PSF against wavelength (top) and against numerical aperture (bottom), all panels on the same physical axis. Panel (c) rescales all eight to rNA/λ ; see the text for why this collapse validates the code rather than the physics.

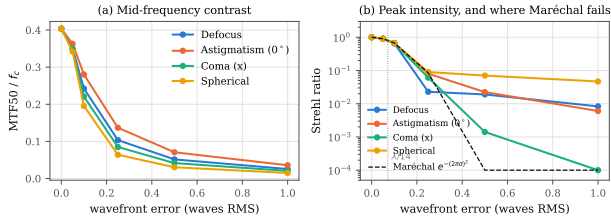


Figure 4: (a) MTF50 against wavefront error. (b) Strehl ratio, with the Maréchal approximation. Two panels rather than one with two vertical scales: a shared frame would invite the reader to read a crossing point that has no meaning.

outside the diffraction-limited regime.

3. MTF50 falls by 39.9% at a tenth of a wave RMS of defocus, a level at which the Strehl ratio (0.6632) still reads as acceptable. Scalar quality metrics and transfer-function metrics are not interchangeable.
4. Regularization strength matters more than method choice, and PSNR and SSIM select different settings. Reporting a single metric would have produced a different recommendation.
5. For Richardson–Lucy the iteration count *is* the regularizer: the optimum moves from 60 iterations at zero noise to 3 at $\sigma = 0.05$.
6. Deconvolution’s advantage is not robust to PSF error: a 0.25-wave error in the assumed defocus costs 22.85 dB $\rightarrow$ 20.69 dB.
7. Undersampling corrupts rather than degrades, and

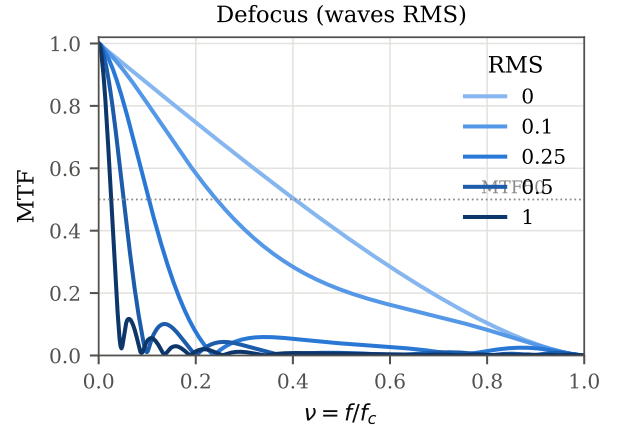


Figure 5: Radial MTF at five defocus severities. The ramp runs light to dark with increasing aberration, so the ordering is readable without the legend.

how much depends on the object’s spectrum, not only on the detector.

9 Limitations

Scalar diffraction theory is assumed throughout: no polarization, and the paraxial-to-moderate-NA regime. High-NA vectorial effects are outside the model and the NA 1.0 rows should be read with that in mind. The imaging is shift-invariant, so field-dependent aberration is not modelled. Deconvolution is given the true PSF except where stated. Richardson–Lucy requires non-negative data, so Gaussian-noise inputs are clipped

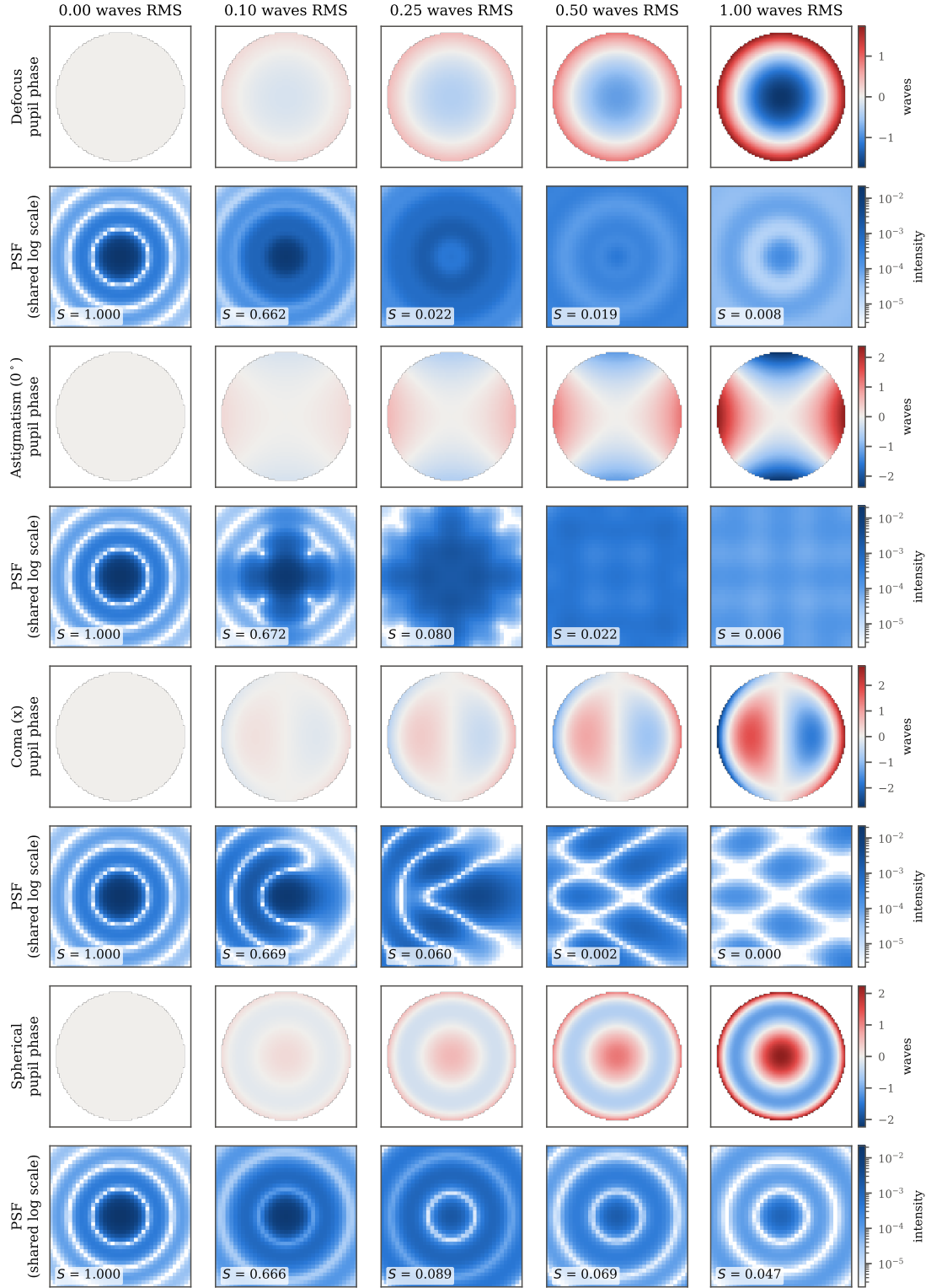


Figure 6: Pupil phase (diverging scale, shared within each row) and the resulting PSF (shared logarithmic scale across the whole figure) for four aberrations at five severities. Both scaling choices are deliberate: per-panel auto-scaling makes a tenth-wave and a one-wave aberration look identical, which hides the only variable being swept.

Table 3: Optical metrics against aberration type and severity. Strehl is the on-axis intensity ratio, which equals the pupil-integral form $|\langle e^{i\phi} \rangle|^2$ exactly; the Maréchal column is the small-aberration approximation to the same quantity.

Aberration	RMS (waves)	PV (waves)	Strehl	Maréchal	MTF50/ f_c	MTF10/ f_c	FWHM (nm)
Defocus	0.00	0.00	1.0000	1.0000	0.4037	0.8049	547.7
Defocus	0.05	0.17	0.9053	0.9060	0.3497	0.7971	549.3
Defocus	0.10	0.35	0.6632	0.6738	0.2425	0.7597	555.4
Defocus	0.25	0.87	0.0229	0.0848	0.1034	0.1911	—
Defocus	0.50	1.73	0.0190	0.0001	0.0515	0.0856	765.7
Defocus	1.00	3.46	0.0083	0.0000	0.0256	0.0407	567.1
Astigmatism	0.00	0.00	1.0000	1.0000	0.4037	0.8049	547.7
Astigmatism	0.05	0.24	0.9063	0.9060	0.3629	0.7869	557.0
Astigmatism	0.10	0.49	0.6727	0.6738	0.2802	0.7030	591.4
Astigmatism	0.25	1.22	0.0803	0.0848	0.1369	0.2557	1892.8
Astigmatism	0.50	2.45	0.0224	0.0001	0.0707	0.1183	4255.2
Astigmatism	1.00	4.90	0.0060	0.0000	0.0357	0.0571	9045.7
Coma	0.00	0.00	1.0000	1.0000	0.4037	0.8049	547.7
Coma	0.05	0.28	0.9060	0.9060	0.3479	0.8009	548.6
Coma	0.10	0.57	0.6702	0.6738	0.2208	0.7863	551.4
Coma	0.25	1.41	0.0610	0.0848	0.0850	0.4634	1983.2
Coma	0.50	2.83	0.0014	0.0001	0.0415	0.2155	—
Coma	1.00	5.66	0.0000	0.0000	0.0206	0.1063	—
Spherical	0.00	0.00	1.0000	1.0000	0.4037	0.8049	547.7
Spherical	0.05	0.17	0.9058	0.9060	0.3416	0.7857	547.4
Spherical	0.10	0.34	0.6676	0.6738	0.1954	0.7383	546.3
Spherical	0.25	0.84	0.0899	0.0848	0.0640	0.1462	532.5
Spherical	0.50	1.68	0.0704	0.0001	0.0303	0.0758	537.6
Spherical	1.00	3.35	0.0468	0.0000	0.0149	0.0366	542.5

at zero before it runs, which is a small positive bias in dark regions. The pupil is sampled on a Cartesian grid, so its rim is pixelated; this is the dominant error source in the MTF near cutoff and is why the analytic comparison is reported to $\nu \leq 0.95$. Noise is applied after, not during, detector integration.

10 Reproducibility

Every experiment is driven by a YAML configuration file, uses a seed derived deterministically from a label rather than from a shared random stream, and writes its numerical results as CSV and JSON alongside a provenance record of library versions and platform. The figures are generated programmatically, the LaTeX tables are generated from the CSVs, and the numbers quoted in this prose are LaTeX macros written from the same CSVs — so a changed parameter changes the text, and a deleted result makes this document fail to build rather than print a stale value.

References

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Table 4: PSNR (dB) against noise condition and restoration method. The restoration is given the exact PSF in every case, so these are upper bounds; Section 6.1 measures what that assumption is worth.

Condition	No restoration	Naive inverse	Wiener (constant K)	Wiener (spectral K)	Richardson–Lucy
$\sigma = 0$	26.08	29.13	30.36	30.36	29.03
$\sigma = 0.01$	25.91	28.31	26.94	28.65	28.45
$\sigma = 0.025$	25.10	25.56	25.66	27.57	26.23
$\sigma = 0.05$	23.10	21.43	24.46	26.41	22.48
$\sigma = 0.1$	19.32	16.47	20.92	24.21	17.62
100 photons	21.29	18.83	23.03	25.52	19.96
500 photons	24.58	24.28	24.60	27.08	25.02
1000 photons	25.27	26.00	25.09	27.62	26.56
5000 photons	25.91	28.29	26.34	28.58	28.42
10000 photons	25.99	28.69	26.78	28.88	28.71

Table 5: Every method across its own regularization parameter, at $\sigma = 0.02$. Runtime is wall-clock for one 512^2 image.

Method	Parameter	PSNR (dB)	SSIM	Runtime (ms)
No restoration	-	25.44	0.605	0.0
Naive inverse	—H— ζ 0.001	7.24	0.022	31.7
Naive inverse	—H— ζ 0.01	15.94	0.139	31.6
Naive inverse	—H— ζ 0.05	24.00	0.398	31.7
Naive inverse	—H— ζ 0.1	26.52	0.560	31.9
Naive inverse	—H— ζ 0.2	27.48	0.690	31.7
Wiener (constant K)	$K=1e-05$	11.24	0.065	32.1
Wiener (constant K)	$K=0.0001$	16.70	0.155	31.3
Wiener (constant K)	$K=0.001$	22.58	0.331	31.5
Wiener (constant K)	$K=0.01$	26.95	0.593	30.8
Wiener (constant K)	$K=0.1$	23.27	0.751	42.0
Wiener (spectral K)	estimated P_n/P_f	27.89	0.714	47.9
Richardson–Lucy	5 iterations	27.01	0.634	193.2
Richardson–Lucy	10 iterations	27.15	0.623	397.8
Richardson–Lucy	20 iterations	27.05	0.599	738.5
Richardson–Lucy	50 iterations	26.46	0.543	1846.8

Table 6: Where Richardson–Lucy peaks. The optimum moves by an order of magnitude with the noise level, and PSNR and SSIM do not agree on it.

Gaussian σ	best-PSNR iter.	PSNR (dB)	best-SSIM iter.	SSIM	PSNR at 60	SSIM at 60
0	60	29.46	60	0.855	29.46	0.855
0.01	34	28.50	9	0.761	28.43	0.721
0.05	3	23.57	3	0.355	20.66	0.251

Table 7: Restoration quality when the assumed defocus differs from the true one (0.25 waves RMS).

Defocus error (waves)	Richardson–Lucy PSNR	Wiener (constant K) PSNR	Richardson–Lucy SSIM	Wiener (constant K) SSIM
-0.25	21.18	20.69	0.505	0.320
-0.15	21.18	20.45	0.498	0.278
-0.10	21.43	19.48	0.519	0.181
-0.05	22.14	20.47	0.562	0.252
0.00	22.51	22.85	0.578	0.339
0.05	21.90	20.58	0.560	0.314
0.10	20.79	16.34	0.528	0.180
0.15	19.77	13.89	0.502	0.094
0.25	18.23	11.42	0.477	0.145

Table 8: Detector pitch as a multiple of the Nyquist pitch $\lambda/(4NA) = 0.2660 \mu\text{m}$, with the fraction of modulation that folds and the pixel MTF at the optical cutoff.

Object	pitch/ p_{Nyq}	pitch (μm)	folded (%)	pixel MTF at f_c	SSIM (integrating)	SSIM (point)
camera	0.50	0.133	0.00	0.900	0.879	0.879
camera	1.00	0.266	0.06	0.637	0.922	0.918
camera	1.50	0.399	0.11	0.300	0.949	0.927
camera	2.00	0.532	0.20	0.000	0.967	0.920
camera	3.00	0.798	0.54	0.212	0.981	0.886
camera	4.00	1.064	1.09	0.000	0.988	0.860
frequency_sweep	0.50	0.133	0.00	0.900	0.529	0.529
frequency_sweep	1.00	0.266	0.09	0.637	0.553	0.545
frequency_sweep	1.50	0.399	1.65	0.300	0.555	0.497
frequency_sweep	2.00	0.532	7.98	0.000	0.704	0.455
frequency_sweep	3.00	0.798	23.02	0.212	0.665	0.409
frequency_sweep	4.00	1.064	36.72	0.000	0.724	0.413
siemens_star	0.50	0.133	0.00	0.900	0.650	0.650
siemens_star	1.00	0.266	0.00	0.637	0.748	0.746
siemens_star	1.50	0.399	0.03	0.300	0.828	0.808
siemens_star	2.00	0.532	0.20	0.000	0.888	0.843
siemens_star	3.00	0.798	1.08	0.212	0.965	0.874
siemens_star	4.00	1.064	2.47	0.000	0.991	0.874

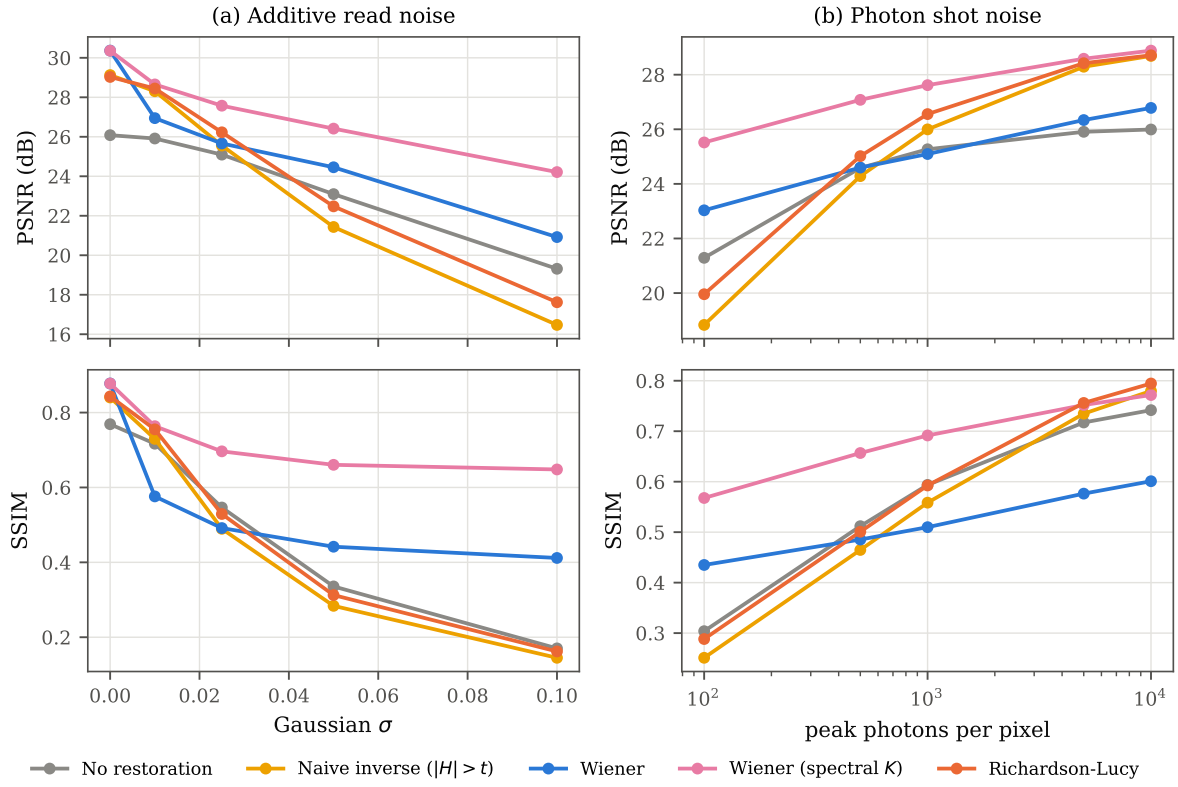


Figure 7: PSNR and SSIM against Gaussian σ and against peak photon count, for five treatments. Note that a method can gain PSNR while losing SSIM.

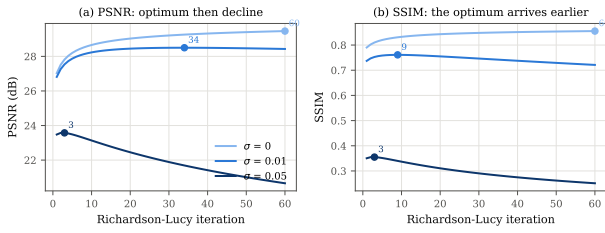


Figure 8: Metric against Richardson-Lucy iteration at three noise levels, with each optimum marked. Maximum likelihood applied to noisy data eventually fits the noise: the iteration does not converge to something good and stop.

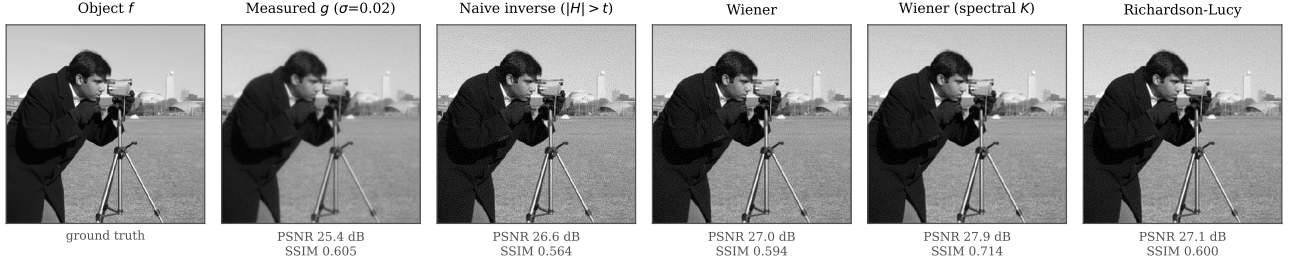


Figure 9: Object, measurement and four restorations at $\sigma = 0.02$, each captioned with its own measured metrics so the visual impression and the numbers cannot drift apart.

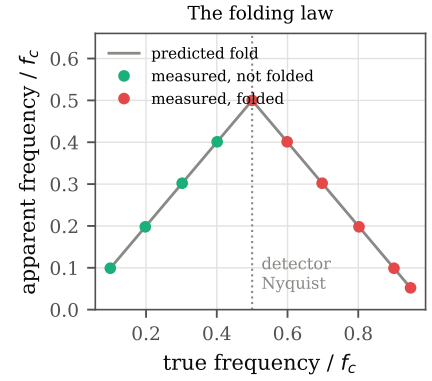
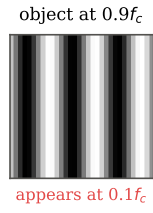
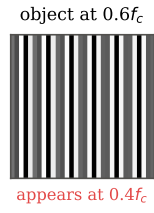
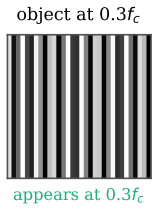


Figure 10: Aliasing as a measurement. Left: three gratings through identical optics and an identical detector; the last two are above the detector's Nyquist frequency and return as coarser patterns. Right: apparent against true frequency, measured and predicted.